

1. What is the difference between a compiler and an interpreter?
2. What is the purpose of the `main()` function?
3. Why do we include header files like `stdio.h` or `iostream`?
4. What is the difference between `gcc` and `g++`?
5. What is an executable file?
6. What happens during compilation?
7. What is the difference between compile-time and run-time errors?
8. What is the purpose of the `return 0;` statement?
9. Why is C called a procedural programming language?
10. What are the advantages of using an IDE over a command-line compiler?

Find Ans for below Questions

1. Is every program an algorithm?
2. Is every algorithm a program?
3. Can an algorithm exist without a programming language?
4. Can there be more than one correct algorithm for the same problem?
5. Is pseudocode executable?
6. Is a flowchart mandatory before coding?
7. Can a program compile successfully but still produce incorrect output?
8. Does compilation guarantee that the program is correct?
9. Can an algorithm have infinite steps?
10. Can a flowchart have multiple end points?
11. Why do computers understand machine language but not C?
12. Why can't humans directly write machine code for large software?
13. Which takes more memory: source code or executable?
14. Is faster execution always better?
15. Which is more important: readability or optimization?

Write Code (hello.c)

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Save (Ctrl + S)

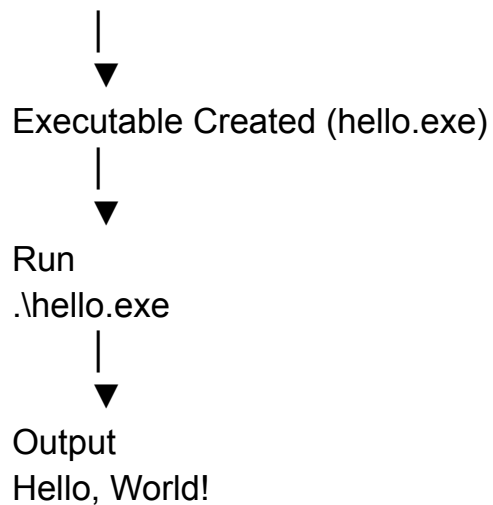
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Compile

`gcc hello.c -o hello`

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Stage	Tool	Why?
Beginner	Code::Blocks	Easy setup with built-in compiler
Beginner → Intermediate	Command Prompt (CLI)	Learn compilation process
Intermediate	VS Code	Industry-standard workflow
Advanced C++	Visual Studio / CLion	Large projects and debugging

Feature	CLI	VS Code	Code::Blocks	Visual Studio	Online
Installation	GCC	VS Code + GCC	IDE + GCC	IDE	None
Debugging	Limited	Yes	Yes	Excellent	No
IntelliSense	No	Yes	Basic	Excellent	Limited
Build & Run	Manual	Manual/Tasks	One Click	One Click	One Click
Beginner Friendly	Medium	High	Very High	Medium	Very High

Industry Usage	High	Very High	Medium	High	Low
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Steps for preparing your System environment

Step	Activity	Action	Expected Result
1	Install VS Code	Download and install VS Code from https://code.visualstudio.com	VS Code installed
2	Install GCC Compiler	Install MSYS2/MinGW-w64	GCC/G++ compiler installed
3	Verify Compiler	Open Command Prompt and run gcc --version and g++ --version	Compiler version displayed
4	Open VS Code	Launch Visual Studio Code	VS Code opens successfully
5	Install C/C++ Extension	Press Ctrl + Shift + X → Search C/C++ → Install	C/C++ support enabled
6	Create Project Folder	Create a folder (e.g., C_Programs) and open it in VS Code	Workspace ready
7	Create Program	Create a file named hello.c or hello.cpp	Source file created
8	Write & Save Code	Type your program and press Ctrl + S	Program saved
9	Open Terminal	Press Ctrl + `	Integrated terminal opens
10	Compile Program	C: gcc hello.c -o hello C++: g++ hello.cpp -o hello	Executable created
11	Run Program	Type hello or hello.exe	Program output displayed

12	Debug (Optional)	Press F5	Program runs in debug mode
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Program No.	Program Name	Problem Statement	Concepts Covered	Sample Input	Expected Output
1	Hello World	Write a program to display "Hello, World!" on the screen.	main(), printf()/cout, Header Files	No Input	Hello, World!
2	Welcome Message	Accept the user's name and display a personalized welcome message.	Variables, Input/Output (scanf()/cin)	Parth	Welcome, Parth!
3	Sum of Two Numbers	Accept two integers from the user and display their sum.	Variables, Arithmetic Operators	10, 20	Sum = 30
4	Simple Calculator	Accept two numbers and perform Addition, Subtraction, Multiplication, and Division.	Operators, Input/Output, Arithmetic Expressions	20, 5	Addition=25 Subtraction=15 Multiplication=100 Division=4
5	Student Marks Calculator	Accept marks of 5 subjects and calculate Total, Average, and Percentage.	Variables, Arithmetic Operations, Formatted Output	80, 75, 90, 85, 70	Total = 400 Average = 80 Percentage = 80%

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https://github.com/ParthShuklaa/ST_GNC_JULY2026_DSA.git